

Concerns over power shortage call for a careful review of India's energy mix

Coal production in the country has now crossed a record 1 billion tonnes — this is **in line with** the action plan of the coal ministry, which had targeted an output of 1,080 million tonnes in 2024-25. Data from the ministry shows that coal production touched 997.83 million tonnes in 2023-24, up 11.7 per cent from 893.19 million tonnes in 2022-23. A **substantial** part of this is **accounted for** by Coal India and its **subsidiaries**. Production by Coal India had increased to 773.81 million tonnes in 2023-24, up from 703.20 million tonnes in 2022-23. Despite this record production, however, **those** managing the electricity grids **are** worried as summer **approaches**.

The problem, in part, **stems from** how India's energy mix has evolved in the recent past. Thermal power **dominates** India's baseload power capacity. In recent years, however, **additions** to thermal capacity **have** been **muted** — the installed capacity rose from 230 GW in 2019-20 to just 243 GW in 2023-24. **Much** of the increase in installed power capacity during these years **has** occurred in the renewables space — the installed capacity rose from 88.26 GW in 2019-20 to 143.64 GW in 2023-24 (renewables include solar, wind, bio-power and small hydro power). However, despite this, renewables still account for just 13 per cent of all-India energy generation. During the summer months, the problem is **acute** as solar generation **tapers off** by the evenings while household demand remains high and storage capacities are limited. **In order to** fill this gap, baseload capacities are **stepped up**, but in the absence of an expansion in such capacities, there is a shortfall in supply. As per reports, electricity demand is expected to **peak** at 273 GW in June 2025 compared to a minimum of 135 GW in December-January. "Shortages are more **likely** to occur during non-solar hours in May, June, July and August 2025," says a report from the National Load Despatch Centre with the months of May and June identified as "high-risk months for shortages".

The National Electricity Plan (2017-22) had focused more on **ramping up** renewable capacities in the country. As per its **assessment**, the expected renewable (solar, wind, biomass and small hydro) capacity **addition** between 2017 and 2022 **was** **pegged at** 117.7 GW. However, the **concerns** that are being raised **call for** carrying out a more careful assessment of the country's energy mix — thermal, nuclear and renewables. As the economy continues its **forward march**, as electricity demand continues to rise — peak electricity demand is expected to touch 298.7 GW by 2026-27 — policy must ensure a **steady** supply of power.

[Practice Exercise]

- Red/blue coloring of words in the sentence indicates subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'.

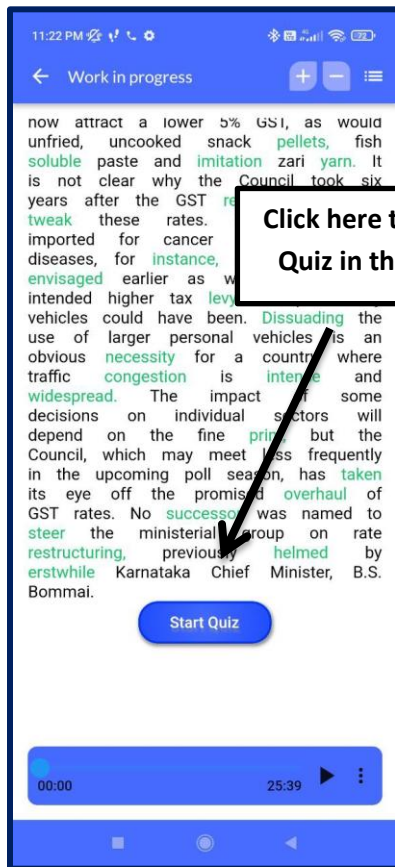
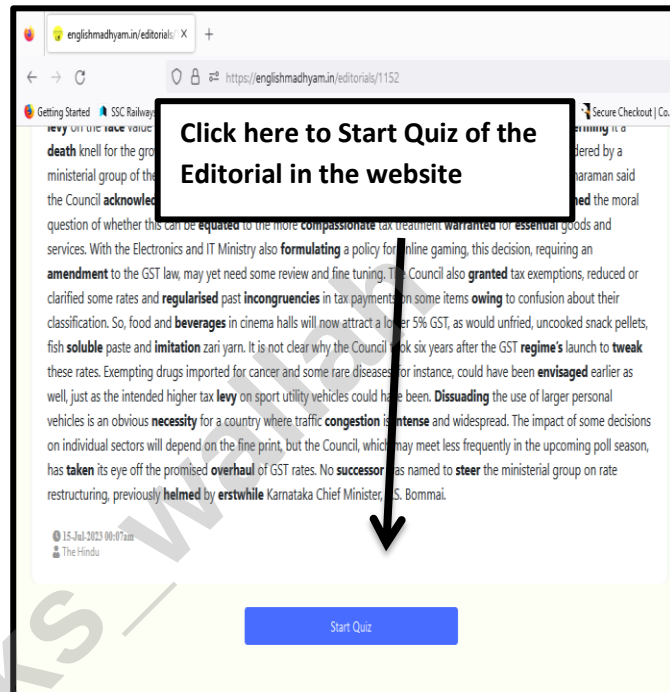
Vocabulary

1. **Call for** (phrasal verb) – Demand, Require, Necessitate, Ask for, Request मांग करना
2. **In line with** (phrase) – Consistent with, In accordance with, Aligned with, Following, Conforming to के अनुरूप
3. **Substantial** (adjective) – Significant, Considerable, Large, Ample, Sizeable पर्याप्त
4. **Account for** (phrasal verb) – Constitute, comprise, make up, form, total बनाना
5. **Subsidiary** (noun) – Branch, Division, Affiliate, Offshoot, Wing सहायक कंपनी
6. **Approach** (verb) – Near, Come close, Advance, Draw near, Move towards नजदीक आना
7. **Stem from** (phrasal verb) – Originate from, Arise from, Result from, Emerge from, Be caused by उत्पन्न होना
8. **Dominate** (verb) – Control, Rule, Prevail, Overpower, Overshadow हावी होना
9. **Muted** (adjective) – Subdued, Limited, Restrained, Softened, Dampened मंद
10. **Acute** (adjective) – Severe, Intense, Critical, Sharp, Serious तीव्र
11. **Taper off** (phrasal verb) – Decrease, Decline, Diminish, Reduce, Subside कम होना
12. **Step up** (phrasal verb) – Increase, Boost, Enhance, Intensify, Accelerate बढ़ाना
13. **Peak** (verb) – Reach the highest point, Climax, Top, Culminate, Max out चरम पर पहुंचना
14. **Likely** (adjective) – Probable, Possible, Expected, Plausible, Anticipated संभावित
15. **Ramp up** (phrasal verb) – Increase, Expand, Boost, Scale up, Enhance तेजी से बढ़ाना
16. **Assessment** (noun) – Evaluation, Analysis, Appraisal, Judgment, Review मूल्यांकन
17. **Peg (at)** (verb) – Fix, Set, Anchor, Stabilize, Lock निर्धारित करना
18. **Forward march** (noun) – Progress, Advancement, Development, Growth, Movement प्रगति
19. **Steady** (adjective) – Stable, Consistent, Constant, Reliable, Unchanging स्थिर

Summary of the Editorial

1. **Record coal production:** India's coal production surpassed 1 billion tonnes in 2023-24, with Coal India and its subsidiaries contributing significantly to this achievement.
2. **Coal ministry's target:** The coal ministry aims to achieve an output of 1,080 million tonnes by 2024-25, reflecting a focus on increasing domestic coal production.
3. **Thermal power dominance:** Thermal power remains the backbone of India's baseload power capacity, but its growth has been slow, rising only from 230 GW in 2019-20 to 243 GW in 2023-24.
4. **Renewable energy growth:** Renewable energy capacity has increased significantly, from 88.26 GW in 2019-20 to 143.64 GW in 2023-24, driven by solar, wind, bio-power, and small hydro projects.
5. **Limited contribution of renewables:** Despite growth, renewables account for only 13% of India's total energy generation, highlighting their limited role in meeting baseload demand.
6. **Summer power challenges:** Solar generation declines in the evenings during summer, while household demand remains high, creating a supply-demand gap.
7. **Storage limitations:** Limited energy storage capacity exacerbates the challenge of integrating renewables into the grid during peak demand periods.
8. **Rising electricity demand:** Peak electricity demand is expected to reach 273 GW in June 2025, compared to a minimum of 135 GW in winter months, straining the power grid.
9. **High-risk months:** May and June 2025 are identified as high-risk months for power shortages, particularly during non-solar hours.
10. **Baseload capacity shortfall:** Insufficient expansion in baseload thermal capacity has led to concerns about meeting rising electricity demand.
11. **National Electricity Plan:** The 2017-22 plan emphasized renewable energy, targeting 117.7 GW of renewable capacity addition, but overlooked the need for a balanced energy mix.
12. **Need for energy mix review:** Current concerns call for a reassessment of India's energy mix, including thermal, nuclear, and renewable sources, to ensure reliable power supply.
13. **Economic growth and power demand:** As India's economy grows, peak electricity demand is projected to reach 298.7 GW by 2026-27, necessitating robust energy planning.
14. **Policy focus:** Policymakers must prioritize a steady power supply by balancing renewable energy growth with the expansion of baseload thermal and nuclear capacities.
15. **Strategic planning:** A comprehensive review of India's energy strategy is essential to address power shortages and ensure sustainable energy security.

Practice Exercise: Banking Pattern Based

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